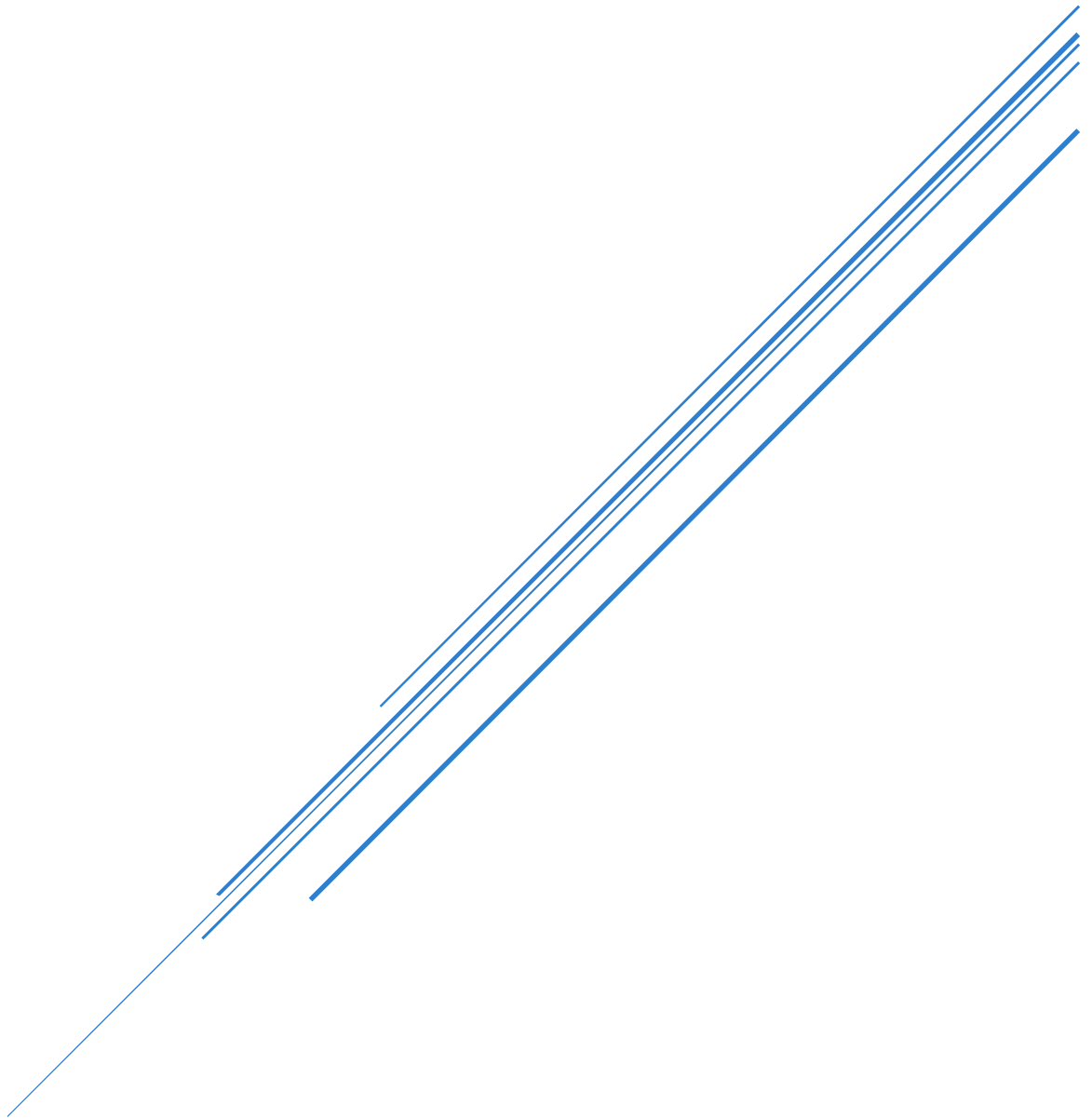


ASSESSMENT COMPTIA A+

A+ - Troubleshooting and Helpdesk



❖ Section A — Concept Application

SCENARIO-1

- You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.
- **Question:** Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together



When selecting hardware for a new office workstation, I would check the compatibility of the motherboard, CPU, RAM, and storage device before purchasing.

1. **Motherboard:** First, I would select a motherboard with the required CPU socket and support for the required RAM type and storage interfaces such as SATA or M.2.
2. **CPU:** I would choose a CPU that uses the same socket type as the motherboard. I would also check the motherboard's CPU compatibility list and BIOS support.
3. **RAM:** I would select RAM that is supported by the motherboard, such as the correct DDR generation, speed, and capacity. I would also check the number of available RAM slots.
4. **Storage:** I would choose an SSD or NVMe SSD depending on the motherboard's supported storage interfaces. I would check whether the motherboard has the required SATA port or M.2 slot.
5. **Confirm Compatibility:** Before buying, I would check the motherboard manufacturer's specifications and CPU/RAM/storage compatibility information. I would also verify that the CPU socket, RAM type, and storage interface match the motherboard. After assembly, I would check the BIOS/UEFI to confirm that the CPU, RAM, and storage device are detected correctly.

SCENARIO-2

- You are advising a client whose employees mainly run office and productivity applications on their workstations.
- **QUESTION:** Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.



- **HDD (Hard Disk Drive):**
HDDs use magnetic disks and moving mechanical parts to store data. They are generally cheaper and provide large storage capacity, but they are slower because of the mechanical movement.
- **SSD (Solid State Drive):**
SSDs use flash memory and have no moving parts. They are much faster and more reliable than HDDs, which results in quicker boot times and faster application loading.
- **NVMe SSD:**
NVMe is a type of SSD that uses the **PCIe interface** to communicate with the computer. It provides much higher data transfer speeds and lower latency than traditional SATA SSDs.
- **Recommendation:**
For office and productivity workstations, I would recommend an **SSD, preferably an NVMe SSD**. It provides fast boot times, quick application loading, and responsive performance. NVMe also offers higher performance than an HDD and traditional SATA SSD, making it a good choice for modern office computers.

SCENARIO-3

- You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.
- **QUESTION:** Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.



- **GPU (Graphics Processing Unit):**

The GPU handles graphics and helps the computer run design software smoothly. A good GPU can process images, videos, and 3D graphics faster and reduces the workload on the CPU.

- **Cooling System:**

The cooling system removes heat from the CPU and GPU. Since the workstation will be used for several hours, proper cooling prevents overheating and helps maintain stable performance.

- **Conclusion:**

A suitable GPU provides better graphics performance, while a good cooling system keeps the components at a safe temperature during long working hours.

SCENARIO-4

- You are assembling a PC from components sourced from three different vendors.
- **QUESTION: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.**



- Hardware compatibility is important because all computer components must work together properly. When using parts from different vendors, I would check that the motherboard supports the CPU, RAM, storage, and other components.
- If components are incompatible, problems may occur such as the computer not starting, RAM or storage not being detected, poor performance, or system instability. In some cases, the component may not fit physically or may require a different connection or specification.
- I would check the specifications of each component before assembling the PC to make sure all parts are compatible.

SCENARIO-5

- You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.
- **QUESTION:** Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.



I would choose a PSU with enough wattage to safely power the motherboard, CPU, GPU, and storage devices. I would also check that the PSU has the correct power connectors for all the components.

Before powering on the PC, I would:

1. Turn off the PSU and unplug the power cable.
2. Check that the motherboard 24-pin power connector is connected properly.
3. Check that the CPU power connector is connected.
4. Make sure the GPU has the required PCIe power connector.
5. Check the storage power and data cables are connected correctly.
6. Check that no loose cables or metal objects are inside the case.
7. Make sure all components are properly installed and take ESD safety precautions.

After checking everything, I would connect the power cable and switch on the PSU to start the computer.

SCENARIO-6

- **You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.**
- **QUESTION: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.**



POST (Power-On Self-Test) checks the main hardware when the computer starts.

To confirm the system passes POST, I would check that:

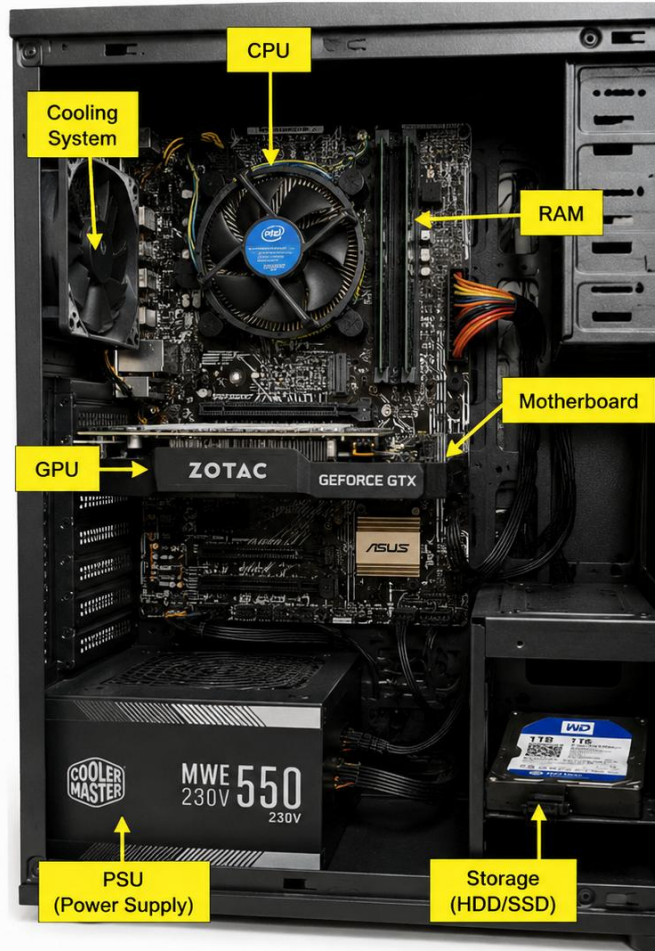
1. The computer powers on normally.
2. The display appears on the monitor.
3. There are no error messages or warning beeps.
4. The BIOS/UEFI screen detects the CPU, RAM, and storage device.
5. The computer continues to the operating system without hardware errors.

If the system does not pass POST, I would turn off the computer and check the power connections, RAM, CPU, and other hardware connections. I would reseat any loose components and check for error messages or beep codes. After fixing the problem, I would power on the system again and check POST.

❖ Section B — Practical Coding Tasks

Task 1:

❖ Component Identification & Functions



Component	Function	Make / Model / Key Specification
 Motherboard	Connects all components together and allows them to communicate. Provides power distribution and supports expansion slots and ports.	ASUS PRIME H610M-E D4 (Micro ATX) - Chipset: Intel H610 - Supports 12th/13th Gen Intel CPUs - RAM: DDR4
 CPU	The brain of the computer. Performs all calculations and processes instructions.	Intel Core i5-12400 6 Cores / 12 Threads - Base Clock: 2.5 GHz - Max Turbo: 4.4 GHz - Socket: LGA1700
 RAM	Temporary memory that stores data currently in use by the system and applications.	Kingston Fury Beast 8GB DDR4 - Capacity: 8GB - Speed: 3200 MHz - Type: DDR4 UDIMM
 GPU	Handles graphics processing and renders images, videos and supports display output.	ZOTAC GeForce GTX 1650 4GB GDDR6 128-bit - PCIe 3.0
 Storage Device	Stores the operating system, applications and all data permanently.	WD Blue 1TB HDD - Capacity: 1TB - Interface: SATA 6Gb/s - Form Factor: 3.5 inch - Speed: 7200 RPM
 Cooling System	Removes heat from the CPU (and other components) to maintain safe temperatures and ensure stable performance.	Intel Stock Cooler - Type: Air Cooler 92mm Fan - For LGA1700 4-Pin Connector
 Power Supply (PSU)	Provides electrical power to all components in the system.	Cooler Master MWE 550 - Wattage: 550W 80+ Bronze - Input: 230V

- Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.
- Note the make/model or key specification of each component where visible.
- Present the findings as a labeled component list or table.

Component	Function	Make/Model / Key Specification
Motherboard	Connects all computer components together and allows them to communicate.	ASUS PRIME H610M-E D4
CPU	Acts as the brain of the computer. It processes instructions and performs calculations.	Intel Core i5-12400
RAM	Temporary memory that stores data currently being used by the computer and applications.	Kingston Fury Beast 8GB DDR4
GPU	Processes graphics, images and videos and sends the output to the monitor.	ZOTAC GeForce GTX 1650 4GB
Storage Device	Stores the operating system, applications, files and other data permanently.	WD Blue 1TB HDD
Cooling System	Removes heat from the CPU and other components and prevents overheating.	Intel Stock Cooler

Task 2:

❖ Motherboard Ports & Socket Identification

- Identify and label all major motherboard ports (SATA, power, USB headers, etc.).

Port / Connector	Function
24-Pin ATX Power	Provides main power to the motherboard.
8-Pin CPU Power	Provides power to the CPU.
SATA Ports	Connect SATA HDDs, SSDs, and optical drives.
M.2 Slot	Used to install an M.2/NVMe SSD.
USB 2.0 Header	Connects front-panel USB 2.0 ports.
USB 3.2 Header	Connects front-panel high-speed USB ports.
Front Panel Header	Connects Power Switch, Reset Switch, Power LED, and HDD LED.
PCIe x16 Slot	Used mainly for a graphics card (GPU).
PCIe x1 Slot	Used for expansion cards such as sound or network cards.
CPU Socket (LGA1700)	Holds the CPU and connects it electrically to the motherboard.
RAM/DIMM Slots	Used to install RAM modules.
Rear USB Ports	Used to connect keyboard, mouse, USB drives, etc.
LAN/RJ-45 Port	Provides wired network/internet connection.
Audio Jacks	Used to connect speakers, headphones, and microphones.
HDMI/VGA	Provides display output when supported by the CPU/motherboard.

- **Identify all expansion slots (PCIe, M.2, etc.) and their purpose.**

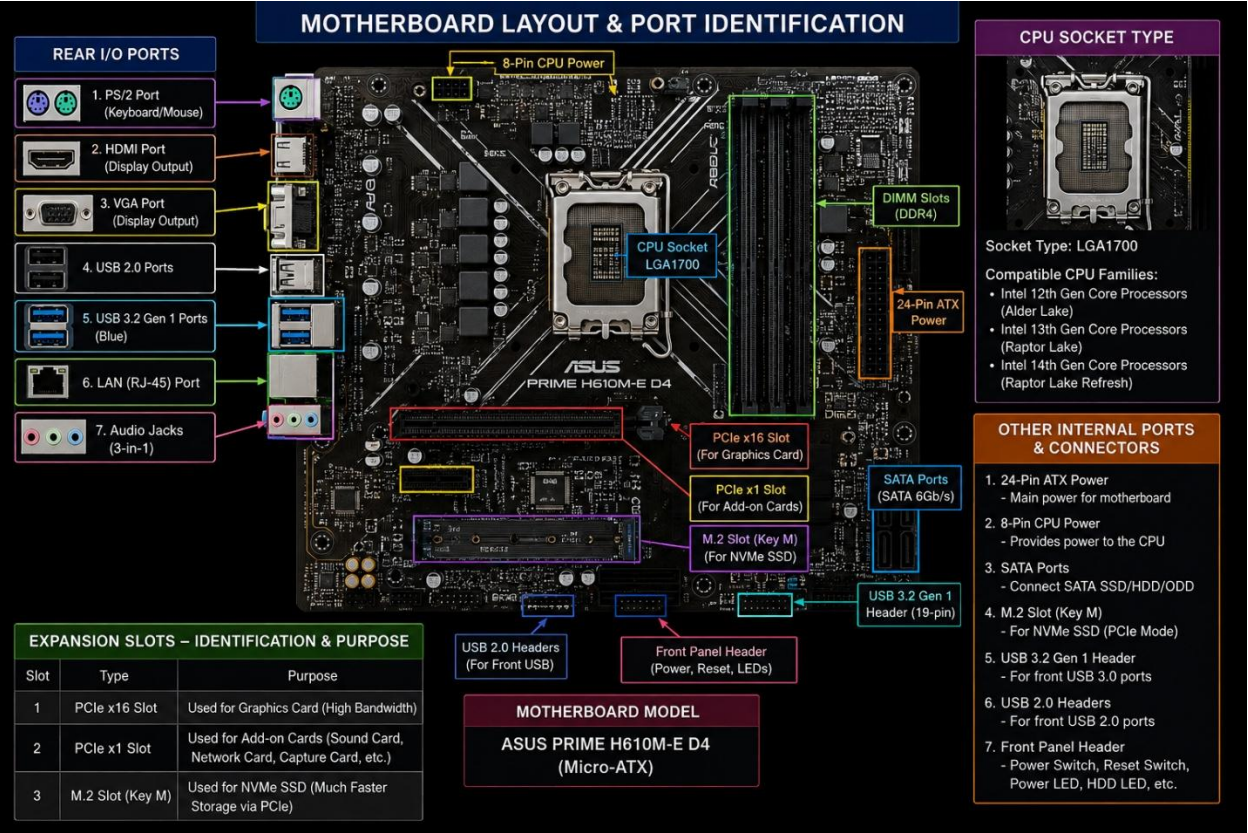
Expansion Slot	Purpose
PCIe x16 Slot	Used mainly for installing a GPU/graphics card. It provides high-speed communication between the GPU and motherboard.
PCIe x1 Slot	Used for smaller expansion cards such as sound cards, network cards, or other add-on cards.
M.2 Slot	Used to install an M.2 SSD, especially an NVMe SSD, for fast storage.

- **Identify the CPU socket type and note which CPU families are compatible with it.**

For the **ASUS PRIME H610M-E D4** motherboard shown in our diagram:

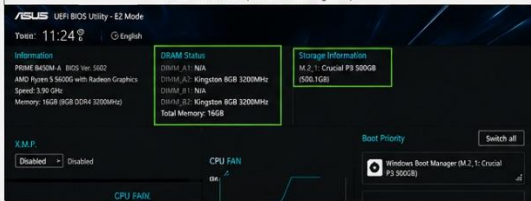
- **CPU Socket:** LGA 1700
- **Compatible CPU families:** Intel 12th Gen, 13th Gen, and 14th Gen Core processors
- **Purpose:** The CPU socket holds the processor and provides the electrical connection between the CPU and motherboard.

- Photograph or diagram the motherboard with labels.



Task 4:

❖ RAM & Storage Installation

1. ESD PRECAUTIONS		2. RAM MODULE REMOVAL			3. RAM MODULE REINSTALLATION		
	- Shut down the PC and switch off the PSU. - Unplug the power cable. - Wear anti-static wrist strap. - Work on a clean, non-conductive surface. - Touch the metal chassis to discharge static before handling components.						
4. NVME SSD REMOVAL				5. NVME SSD REINSTALLATION			
							
CONNECTION TYPE USED		6. CONFIRMATION - BIOS DETECTION		7. CONFIRMATION - WINDOWS DETECTION			
 NVMe SSD (M.2 PCIe) No SATA cable used. Connected directly to the M.2 slot on the motherboard.		 Enter BIOS/UEFI (Press DEL during boot) Time: 11:24 English Information: Intel i9000M 4 BIOS Ver: 0007, AMD Ryzen 5 5600G with Radeon Graphics, Speed: 3.90 GHz, Memory: 16GB (8GB DDR4 3200MHz) DRAM Status: DRAM A1: N/A, DRAM A2: Kingston RGB 3200MHz, DRAM A3: N/A, DRAM A4: Kingston RGB 3200MHz, Total Memory: 16GB Storage Information: M.2_1: Crucial P3 500GB (500GB) X.M.P.: Disabled Disabled CPU FAN: On CPU FAN: On Boot Priority: Windows Boot Manager (M.2_1: Crucial P3 500GB)		 Check in Windows (This PC) Quick access: Desktop, Downloads, Documents, Pictures, This PC, Network Devices and drives (2): Local Disk (C:) 298 GB free of 476 GB, Local Disk (D:) 931 GB free of 931 GB RAM Detected: Installed RAM: 16.0 GB (15.9 GB usable) NVMe SSD Detected: Crucial P3 500GB, Drive is working properly.			

✓ **RESULT:** RAM module and NVMe SSD removed and reinstalled successfully. Both components are detected correctly in BIOS and Windows.

I. Safely remove the RAM module, noting ESD precautions taken:

I shut down the computer and disconnected the power cable. I used an anti-static wrist strap and handled the RAM carefully to prevent ESD damage.

II. Reinstall the RAM module correctly and confirm it is seated properly.:

I aligned the RAM notch with the slot and pressed the RAM down firmly. The side clips locked into place, confirming that the RAM was seated correctly.

III. Remove and reinstall the storage device, noting the connection type used (SATA/NVMe):

I used an NVMe SSD, which is connected directly to the motherboard through the M.2 slot. I removed the screw, took out the SSD, then reinserted it and secured it with the screw.

IV. Confirm both components are detected correctly after reinstallation.

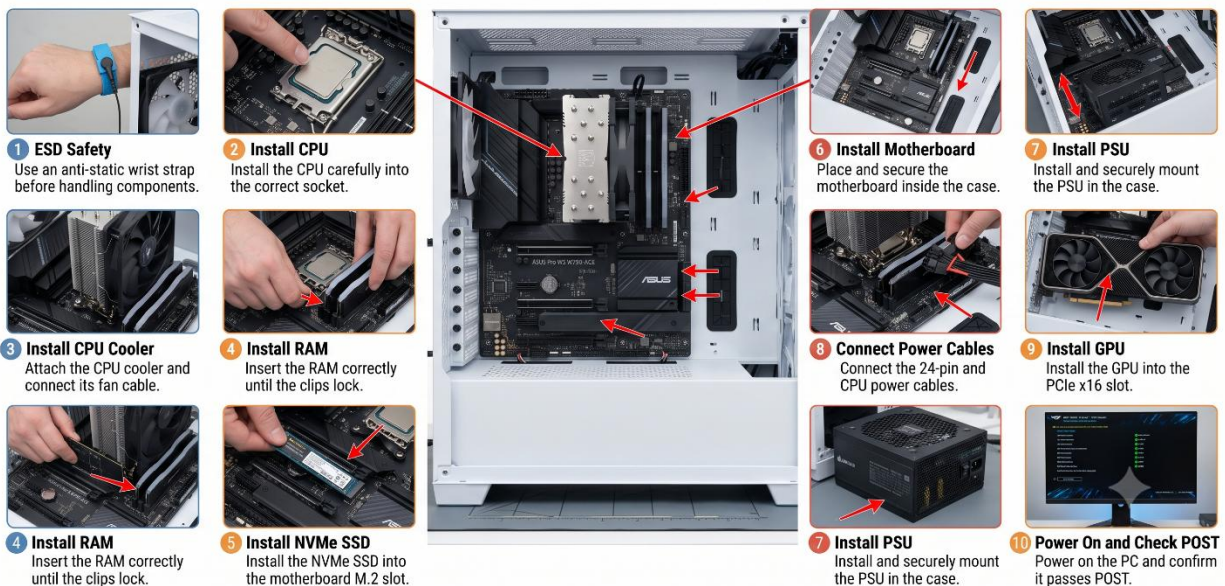
After reinstallation, I started the computer and checked the BIOS/UEFI. The RAM and NVMe SSD were detected correctly, confirming that both components were installed properly.

Task 3:

❖ Full Desktop PC Assembly

- Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.
- Follow correct installation sequence and ESD/safety precautions throughout.
- Document each step with labeled photos or diagrams.
- Power on the system and confirm it passes POST successfully

Task 4: Full Desktop PC Assembly



I. Assembly Sequence:

I followed the correct installation sequence: CPU, CPU cooler, RAM, NVMe SSD, motherboard, PSU, power cables, and GPU. I handled all components carefully and followed ESD safety precautions.

II. Component Installation:

I installed the CPU, RAM, NVMe SSD, GPU, and motherboard in their correct slots. I secured the components properly and connected the required power and data cables.

III. Final Checks:

Before powering on, I checked that all components and cables were properly connected. I also checked that there were no loose screws or cables inside the case.

IV. Power On & POST:

I connected the monitor, keyboard, mouse, and power cable and switched on the system. The BIOS/UEFI screen appeared and detected the hardware, confirming that the system passed POST successfully.

❖ Section C — Mini Capstone Project

❖ Mini Project: Workstation Deployment & Boot-Failure Diagnosis

- **Objective:**

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B

- **Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.**

1. Three Hardware Components to Check First

If a newly assembled PC does not boot, I would first check the PSU, RAM, and motherboard.

- **PSU:** Check whether the PSU is switched on and the motherboard and CPU power cables are connected correctly.
- **RAM:** Check whether the RAM is properly inserted and the locking clips are closed.
- **Motherboard:** Check the motherboard for loose connections, incorrect installation, or any visible problem.

These components are important because a problem with them can prevent the computer from starting or completing POST.

- **Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.**

2. Motherboard–CPU Compatibility

Motherboard and CPU compatibility is important because the CPU must match the motherboard's socket and supported processor family.

In the system I assembled, the ASUS PRIME H610M-E D4 motherboard uses an LGA1700 socket, and the Intel Core i5-12400 is an LGA1700 processor. Therefore, they are compatible and can work together when the required BIOS support is available.

- **Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.**

3. Verifying RAM and Storage Installation

I would check that the RAM is properly seated in the correct DIMM slot and that its clips are locked. After starting the PC, BIOS/UEFI should show the correct RAM capacity.

For the NVMe SSD, I would check that it is correctly inserted into the motherboard's M.2 slot and secured with the screw. BIOS/UEFI should detect the NVMe SSD as a storage device.

- **Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.**

4. Hardware Installation Checklist

To prevent similar failures in future deployments, I would follow these steps:

1. Check compatibility between the motherboard, CPU, RAM, and storage before installation.
2. Use ESD precautions and handle all components carefully.
3. Check all power and data connections before the first boot.
4. Make sure RAM, GPU, and NVMe SSD are properly seated.
5. Check BIOS/UEFI after the first boot to confirm that the hardware is detected.

- **Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.**

5. Why SSD is Better Than HDD for the Office Lab

I would use SSDs instead of HDDs in modern office workstations. SSDs have no moving mechanical parts, so they provide faster boot times, faster application loading, and better overall system responsiveness.

For ABC Solutions' computer lab, employees mainly use office and productivity applications, so fast and reliable storage will make everyday work smoother. SSDs are therefore a better choice for modern office workstations than traditional HDDs.

❖ Section D — AI-Augmented Learning

Section D – AI-Augmented Learning

STEP 1 · BUILD WITH AI

- Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.
- Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.
- Is presented as a clear checklist or simple decision flow a technician could follow on-site.
- Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.

1. Exact Prompt Given to AI

“Create a step-by-step troubleshooting checklist for a junior technician to diagnose a desktop PC that fails to boot immediately after assembly. Put the steps in the correct priority order. Include the motherboard, RAM, storage device, and PSU as possible failure points. Make it a simple checklist that can be followed on-site. Apply the checklist to the ABC Solutions scenario and explain which step could identify the reported issue.”

2. AI's Original Checklist

- **Desktop PC Boot-Failure Troubleshooting Checklist**

1. Check that the PSU is switched on and the power cable is connected.
2. Check the motherboard 24-pin ATX and CPU power connections.
3. Check that the RAM is properly seated in the RAM slot.
4. Check that the CPU and CPU cooler are installed correctly.
5. Check that the GPU is properly seated and connected if required.
6. Check that the storage device is connected correctly.
7. Turn on the PC and check for POST, display output, beep codes, or motherboard error LEDs.
8. Enter BIOS/UEFI and check whether the CPU, RAM, and storage are detected.
9. If the PC still does not boot, reseal the RAM and other components and test again.

STEP 2 · TEST & DEBUG (WITHOUT AI)

- Then, working without AI, review the checklist and find at least one step that is missing, incorrectly ordered, or unsafe (for example, skipping ESD precautions or checking software before physical connections). Fix it yourself. **Corrected Checklist**

1. **Power off and follow ESD precautions before opening or touching any components.**
2. **Check the PSU:** Make sure the PSU switch is ON, the power cable is connected, and the PSU is providing power.
3. **Check motherboard power:** Confirm the 24-pin ATX and CPU power connectors are firmly connected.
4. **Check RAM:** Make sure the RAM is installed in the correct slot and both locking clips are closed.
5. **Check CPU and cooler:** Confirm the CPU is correctly seated and the CPU cooler is securely installed.
6. **Check GPU:** Make sure the GPU is properly seated in the PCIe slot and its power cable is connected if required.
7. **Check storage:** Confirm the SATA or NVMe storage device is correctly installed and connected.
8. **Power on and check POST:** Look for display output, beep codes, or motherboard diagnostic LEDs.
9. **Check BIOS/UEFI:** Confirm that the CPU, RAM, and storage device are detected correctly.
10. **If the problem remains:** Reseat the suspected component and test the system again.

3. What I Changed and Why

- I added an ESD safety check as the first step because the original checklist did not mention it. I also placed physical power and component checks before BIOS/software checks because a newly assembled PC should first be checked for physical installation problems. This makes the troubleshooting process safer and gives the technician a clear priority order.